

DTFT and DFT

Signals and Linear Systems

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Last Time: Four Transforms, One Square

- Last class ended on a promise: one structural rule generates every Fourier-type transform.
- We already know the left column — the CTFS and the CTFT — from continuous-time analysis.
- Today we **derive** the right column: what happens to Fourier's idea when time arrives in samples.

	time is continuous	time is discrete
periodic	Fourier Series (CTFS) known ✓	DFS / DFT today, Part I & III
aperiodic	Fourier Transform (CTFT) known ✓	DTFT today, Part II

- **Part I — From Fourier Series to the DFT:** sample a periodic signal; the Fourier integral becomes a sum; the Discrete Fourier Series (DFS); the DFT and its inverse.
 - Reading the bins: where the negative frequencies live.
- **Part II — The Discrete-Time Fourier Transform:** sample an aperiodic signal; the DTFT, its 2π -periodicity, and its inverse.
- **Part III — The DFT Through the DTFT Lens:** the DFT as N evenly spaced samples of the DTFT; zero-padding; a worked example; bins in hertz.

From Fourier Series to the DFT

- Let's start with a continuous-time periodic signal $x(t)$ with period T_0 :

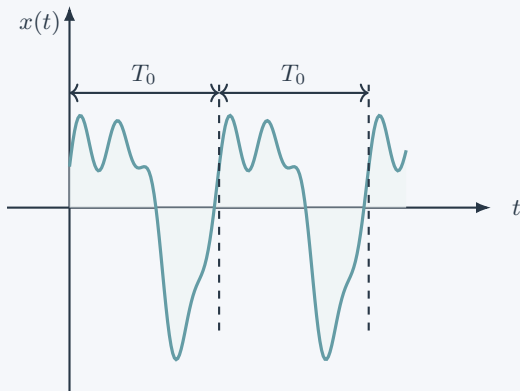
$$x(t + T_0) = x(t), \quad \omega_0 \triangleq \frac{2\pi}{T_0}.$$

- We have learnt the Continuous-Time Fourier Series (CTFS) representation:

CTFS Pair

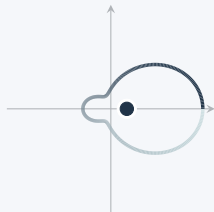
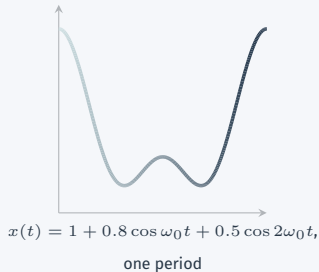
$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$$

$$c_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-jk\omega_0 t} dt$$

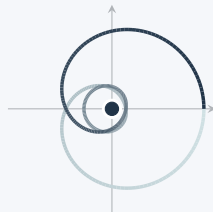


Seeing the Fourier Series: Wind One Period

- Multiply $x(t)$ by $e^{-jk\omega_0 t}$: each value becomes a **radius** and $k\omega_0 t$ an **angle**, so the graph is wound around the origin, k turns per period. The coefficient $c_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-jk\omega_0 t} dt$ is the **center of mass** of the wound curve.



$k = 1$: the loop closes, centroid = $c_1 = 0.4$



$k = 3$: the mass cancels, centroid = $c_3 = 0$

- **One period is enough:** after T_0 the angle has advanced by $2\pi k$, so every later period retraces the same loop. Only the rates $k\omega_0$ close it — hence a line spectrum.

We are interested in a representation for discrete signals. We can sample $x(t)$ to get a discrete

- Sample $x(t)$ uniformly with sampling period T_s .
- Assume T_0 is captured by N samples:

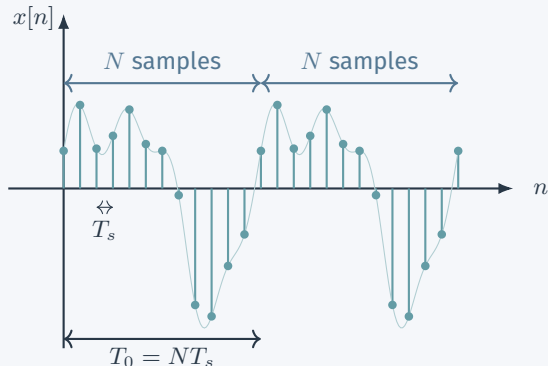
$$T_0 = NT_s \quad (N \in \mathbb{Z}^+).$$

- The purely discrete sequence $x[n]$ is defined as:

$$x[n] \triangleq x(nT_s), \quad n \in \mathbb{Z}.$$

- Thus $x[n]$ is inherently **N -periodic**:

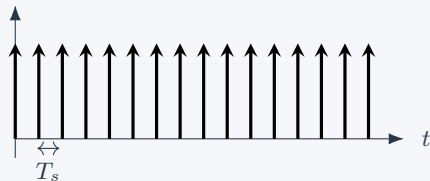
$$x[n + N] = x[n].$$



Sampling as an Impulse Train

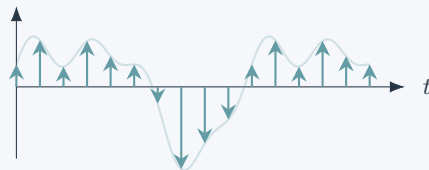
- We can represent the sampled signal mathematically using an impulse train (aka Dirac comb).
- Multiplying the continuous signal $x(t)$ with the impulse train $p(t)$ gives us the sampled signal $x_p(t)$.
- Each sample $x[n]$ is weighted by a Dirac delta function at time nT_s .

$p(t)$



Impulse train $p(t) = \sum \delta(t - nT_s)$.

$x_p(t)$



Sampled signal $x_p(t) = x(t)p(t)$.

$$x_p(t) = \sum_{n=-\infty}^{\infty} x(nT_s) \delta(t - nT_s) = \sum_{n=-\infty}^{\infty} x[n] \delta(t - nT_s).$$

The Integral Becomes a Sum

- The sampled signal $x_p(t)$ is still T_0 -periodic, so it has a CTFS. Let c_k be its coefficients:

$$c_k = \frac{1}{T_0} \int_{T_0} x_p(t) e^{-jk\omega_0 t} dt.$$

- Substitute $x_p(t) = \sum_{n=-\infty}^{\infty} x[n]\delta(t - nT_s)$; the impulses sift the integral:

$$\begin{aligned} c_k &= \frac{1}{T_0} \int_{t_0}^{t_0+T_0} \left(\sum_{n=-\infty}^{\infty} x[n]\delta(t - nT_s) \right) e^{-jk\omega_0 t} dt \\ &= \frac{1}{T_0} \sum_{n=0}^{N-1} x[n] \int_{t_0}^{t_0+T_0} \delta(t - nT_s) e^{-jk\omega_0 t} dt = \frac{1}{T_0} \sum_{n=0}^{N-1} x[n] e^{-jk\omega_0 nT_s} \end{aligned}$$

- Since $\omega_0 T_s = \frac{2\pi}{T_0} T_s = \frac{2\pi}{N}$, we arrive at a **finite sum**:

$$c_k = \frac{1}{T_0} \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N} kn}.$$

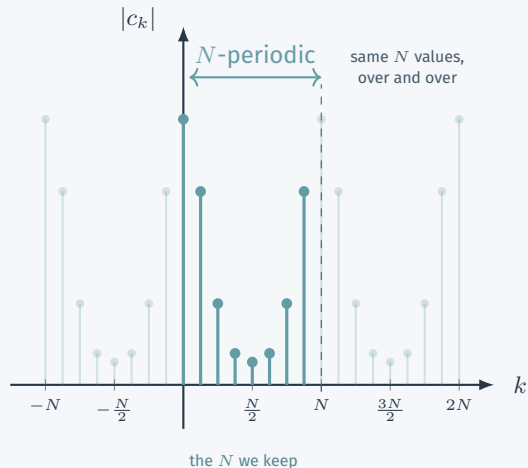
We can take $t_0 = -\frac{T_s}{2}$, so $\delta(t - nT_s)$ is non-zero in $[t_0, t_0 + T_0]$ only for $n = 0, 1, 2, \dots, N-1$.

Only N Distinct Coefficients

- Examine the exponent in c_k :

$$\begin{aligned} c_{k+N} &= \frac{1}{T_0} \sum_{n=0}^{N-1} x[n] e^{-j \frac{2\pi}{N} (k+N)n} \\ &= c_k \underbrace{e^{-j 2\pi n}}_{=1} = c_k \end{aligned}$$

- The frequency coefficients are **N -periodic**: sampling in time made the spectrum periodic!
- Only N unique frequency values** exist before the spectrum repeats.
- We conventionally keep the window $k = 0, 1, \dots, N-1$.



👉 **Interactive:** sample a period, watch a_k repeat with period N

Reading the Bins: Where the Negative Frequencies Went

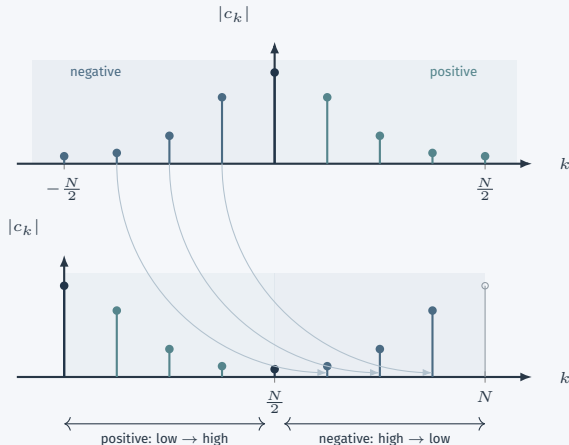
- The natural spectrum is **centered at** $k = 0$: low frequencies in the middle, negative k on the left.
- Periodicity moves each negative coefficient into the window $0, \dots, N - 1$:

$$c_{-k} = c_{N-k}.$$

- The negative side is not lost — it **folds onto the upper bins**.

How to read N bins

Bins $0 \rightarrow \frac{N}{2}$: **positive** frequencies, low $\rightarrow$ high. Bins $\frac{N}{2} \rightarrow N$: **negative** frequencies, high $\rightarrow$ low. A real lowpass signal peaks at $k = 0$ and near $k = N$, and dips at $k = \frac{N}{2}$.



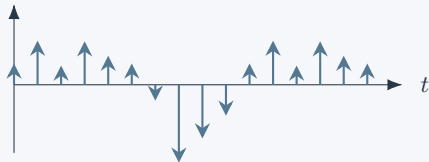
Can CTFS Synthesis Give Us Back $x[n]$? No.

- We cannot simply use the CTFS synthesis equation to recover our discrete signal:

$$\sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t} = x_p(t)$$

- It rebuilds the continuous-time **impulse train** $x_p(t)$ — not the sequence $x[n]$ we want — and it spends infinitely many terms repeating the same N coefficient values.

$x_p(t)$



What CTFS synthesis rebuilds: impulses. ✗

$x[n]$



What we want: the sequence itself. ✓

- But there is a familiar pattern: analysis uses $e^{-j\cdots}$, synthesis uses $e^{+j\cdots}$. Let's build a **discrete synthesis** that uses only the N unique coefficients.

- From the analysis sum, the N unique coefficients are (writing $T_0 c_k$ to clear the scaling):

$$T_0 c_k = \sum_{m=0}^{N-1} x[m] e^{-j \frac{2\pi}{N} km}, \quad k = 0, 1, \dots, N-1.$$

- Multiply both sides by $e^{+j \frac{2\pi}{N} kn}$ and sum over the N unique k :

$$\begin{aligned} \sum_{k=0}^{N-1} (T_0 c_k) e^{j \frac{2\pi}{N} kn} &= \sum_{k=0}^{N-1} \left(\sum_{m=0}^{N-1} x[m] e^{-j \frac{2\pi}{N} km} \right) e^{j \frac{2\pi}{N} kn} \\ &= \sum_{m=0}^{N-1} x[m] \left(\sum_{k=0}^{N-1} e^{j \frac{2\pi}{N} k(n-m)} \right) \end{aligned}$$

- Evaluate the inner summation using the geometric series sum formula. For $n \neq m$:

$$\sum_{k=0}^{N-1} \left(e^{j \frac{2\pi}{N} (n-m)} \right)^k = \frac{1 - e^{j 2\pi (n-m)}}{1 - e^{j \frac{2\pi}{N} (n-m)}} = \frac{1 - 1}{1 - e^{j \frac{2\pi}{N} (n-m)}} = 0$$

- And for $n = m$, the summation simply evaluates to:

$$\sum_{k=0}^{N-1} e^0 = N \implies \sum_{k=0}^{N-1} e^{j \frac{2\pi}{N} k(n-m)} = \begin{cases} N & n = m \\ 0 & n \neq m \end{cases}$$

- All terms in the outer sum vanish except $m = n$:

$$\sum_{k=0}^{N-1} (T_0 c_k) e^{j \frac{2\pi}{N} kn} = x[n] \cdot N \implies x[n] = \frac{1}{N} \sum_{k=0}^{N-1} (T_0 c_k) e^{j \frac{2\pi}{N} kn}.$$

- A **finite sum** recovers the samples exactly — no impulses, no infinite series.
- Notice: both directions only ever touch the N numbers $T_0 c_k$. That combination is the natural discrete-frequency object — **it deserves its own name**.

The Discrete Fourier Series (DFS)

- Forget the continuous signal for a moment: let $\tilde{x}[n]$ be any N -periodic sequence. The same finite sums form a self-contained series representation:

Discrete Fourier Series (DFS) Pair

$$\underbrace{a_k = \frac{1}{N} \sum_{n=0}^{N-1} \tilde{x}[n] e^{-j \frac{2\pi}{N} kn}}_{\text{analysis}}$$

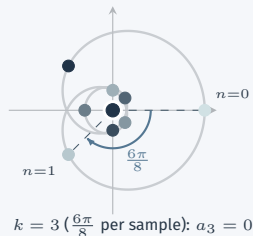
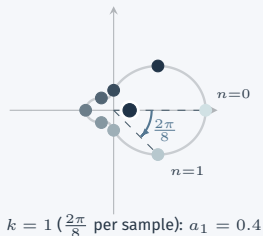
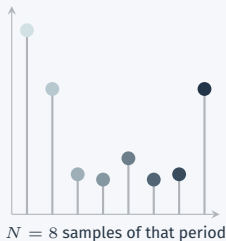
$$\underbrace{\tilde{x}[n] = \sum_{k=0}^{N-1} a_k e^{j \frac{2\pi}{N} kn}}_{\text{synthesis}}$$

- It is the exact discrete analogue of the CTFS — same shape, integral $\rightarrow$ sum — and its validity is exactly the orthogonality trick from the last two slides:

	CTFS (period T_0)	DFS (period N)
Analysis	$c_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-jk \frac{2\pi}{T_0} t} dt$	$a_k = \frac{1}{N} \sum_{n=0}^{N-1} \tilde{x}[n] e^{-jk \frac{2\pi}{N} n}$
Synthesis	$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk \frac{2\pi}{T_0} t}$	$\tilde{x}[n] = \sum_{k=0}^{N-1} a_k e^{jk \frac{2\pi}{N} n}$
Coefficients	infinitely many	N unique ($a_{k+N} = a_k$)

Seeing the DFS: Wind the N Samples

- Wind the N samples instead of the curve: point n sits at radius $x[n]$, angle $-\frac{2\pi}{N}kn$. Their center of mass is the DFS coefficient $a_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$: a plain average — that is why the integral became a sum.



- Every sample turns by the **same angle** $\frac{2\pi}{N}k$; after N samples the points repeat, so one period is still enough. And $k + N$ turns each sample by the same angle mod 2π : only N coefficients are distinct — as derived.

The Discrete Fourier Transform (DFT)

- For a length- N sequence $\{x[n]\}_{n=0}^{N-1}$, define the transform as the **bare analysis sum** (no scale factor). It is customary to write it with the **twiddle factor**

$$W_N \triangleq e^{-j\frac{2\pi}{N}}, \quad W_N^N = 1 \quad (\text{an } N\text{-th root of unity — powers of } W_N \text{ cycle}).$$

DFT / IDFT Pair

$$X[k] \triangleq \sum_{n=0}^{N-1} x[n] W_N^{kn} = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}, \quad k = 0, 1, \dots, N-1$$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn} = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi}{N}kn}, \quad n = 0, 1, \dots, N-1$$

- Connection to the DFS:** $X[k] = N a_k$ — the DFT of a periodic discrete signal is its DFS, up to the $\frac{1}{N}$ convention factor.
- Connection to the sampled continuous signal:** $c_k = \frac{1}{T_0} X[k]$ — the CTFS coefficients of $x_p(t)$ are the DFT values scaled by $\frac{1}{T_0}$.

We Never Needed the Continuous Signal

- Look at the DFT formula again: it consumes only the N numbers $x[0], \dots, x[N-1]$. Nothing in it remembers T_s or $x(t)$.
- The sampling story was **scaffolding**: its only job was to show why the Fourier integral became a sum once the signal lived on samples.
- Any length- N (or N -periodic) list of numbers — an audio buffer, a pixel row, a sensor log — has a DFS/DFT, no continuous ancestor required.

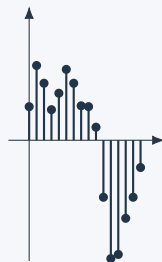


Scaffolding removed

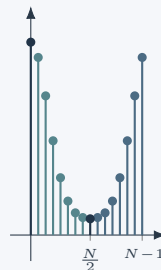
DFS: a Fourier series whose integral collapsed to a sum. DFT: the DFS, rescaled. That is all they are.

- We started with a continuous-time periodic signal $x(t)$.
- Sampling it yielded a **periodic discrete sequence** $x[n]$ with period N .
- Its frequency representation c_k is uniquely defined by exactly N contiguous components.
- The **DFT** transforms exactly N points of $x[n]$ into N unique frequency coefficients $X[k]$.
- The **IDFT** perfectly reconstructs those N points in time from these N frequency values.

N points in Time



N points in Freq



DFT
IDFT

- positive frequencies (low $\rightarrow$ high)
- folded negative frequencies (high $\rightarrow$ low)

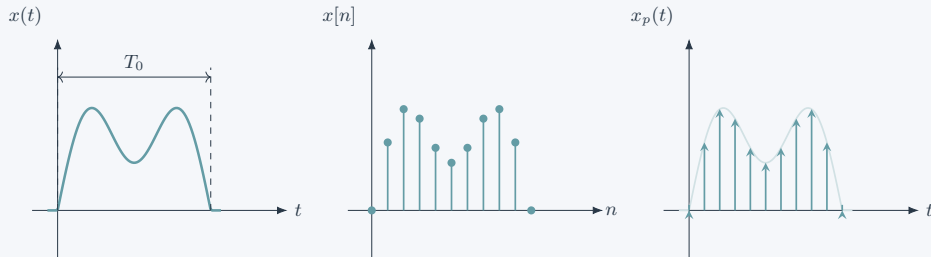
The Discrete-Time Fourier Transform

Now Let the Signal Be Aperiodic

But does this machinery only apply to periodic signals? What if the signal is aperiodic?

- Let's consider an aperiodic signal $x(t)$ restricted to a finite interval T_0 .
- Uniformly sampling this yields a finite-length sequence $x[n]$ that is nonzero only for $0 \leq n \leq N - 1$.
- Using the impulse-train representation, the sampled signal is:

$$x_p(t) = x(t)p(t) = \sum_{n=-\infty}^{\infty} x[n]\delta(t - nT_s) = \sum_{n=0}^{N-1} x[n]\delta(t - nT_s)$$



- An aperiodic signal calls for the Fourier transform. Take the CTFT of the sampled impulse train:

$$X_p(j\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega nT_s}$$

- Now compress the frequency axis by the sampling period, i.e. apply the mapping $\omega T_s \mapsto \omega$.
- Under that relabeling, the same spectrum becomes the **Discrete-Time Fourier Transform (DTFT)**:

Discrete-Time Fourier Transform (DTFT)

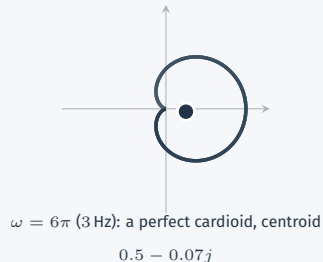
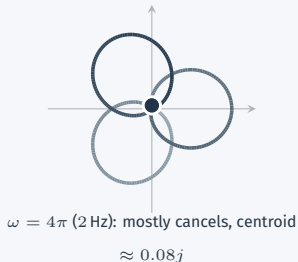
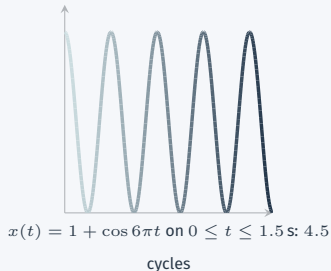
$$X(e^{j\omega}) \triangleq X_p(j\omega)|_{\omega T_s \mapsto \omega} = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}.$$

- Hence, the DTFT is the Fourier transform of the sampled signal, with a compressed frequency axis. The new ω is digital frequency: radians per sample.
- For our length- N capture,

$$X(e^{j\omega}) = \sum_{n=0}^{N-1} x[n]e^{-j\omega n}.$$

Aperiodic: Wind the Whole Segment, at Any Rate

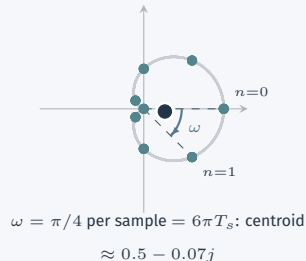
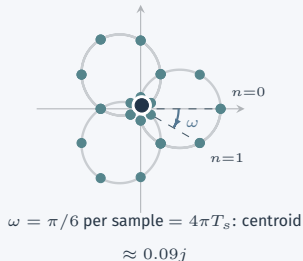
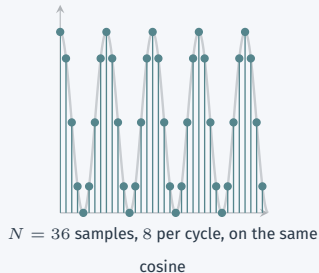
- Take a **finite segment** of a shifted cosine, $x(t) = 1 + \cos 6\pi t$ for $0 \leq t \leq 1.5$ s: nothing repeats outside it, so the winding runs over the **entire** segment, and its centroid times the duration is the transform $X(j\omega) = \int x(t) e^{-j\omega t} dt$. No rate is special, so **every** ω gets its own value: a continuous spectrum.



- At the cosine's own rate every cycle traces the same cardioid $r = 1 + \cos \theta$, and the mass sits to the right of the origin. At any other rate the cycles spread around the circle and nearly cancel; the small residue — here from 4.5 cycles not closing — is the continuous skirt of the spectrum.

Only N Points to Wind: The DTFT and Digital Frequency

- Sample the segment, $x[n] = x(nT_s)$ with $T_s = \frac{1}{24}$ s: the winding places the points $x[n] e^{-j\omega n}$ on the same curve, and with no period all of them count, $X(e^{j\omega}) = \sum_n x[n] e^{-j\omega n} = N \cdot \text{centroid}$. Consecutive points are turned by the **same angle** ω — the digital frequency, in radians per sample, equal to $\omega_{\text{phys}} T_s$.



- 3 Hz becomes $6\pi \cdot \frac{1}{24} = \frac{\pi}{4}$ radians per sample, and the 36 points land on just 8 spots of the cardioid, piling up there. Turning by $\omega + 2\pi$ per sample places the very same points, so $X(e^{j\omega})$ must repeat every 2π — next slide.

The DTFT Is 2π -Periodic

Period of the DTFT

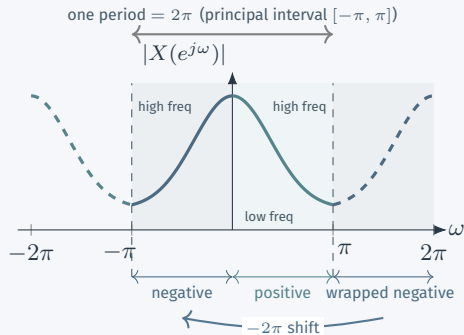
Each term repeats every 2π (as $e^{-j2\pi n} = 1$), so:

$$X(e^{j(\omega+2\pi)}) = X(e^{j\omega})$$

How to Read One Period

- Any 2π -wide window holds one full period.
- A common principal interval is $[-\pi, \pi]$.
- $[\pi, 2\pi]$ is the **folded** copy of $[-\pi, 0]$.

Before $\omega T_s \mapsto \omega$ the axis was physical frequency. Undoing the relabel, $[-\pi, \pi]$ corresponds to $[-\frac{\omega_s}{2}, \frac{\omega_s}{2}]$ with $\omega_s = \frac{2\pi}{T_s}$; we revisit that view in Sampling.



- Same pattern as Part I: synthesis should use $e^{+j\omega n}$. But the spectrum is now a continuous function of ω , so instead of summing over N unique bins we **integrate over one 2π period**:

$$\hat{x}[n] \triangleq \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega.$$

- Substitute the analysis sum $X(e^{j\omega}) = \sum_m x[m] e^{-j\omega m}$ and swap sum and integral:

$$\hat{x}[n] = \sum_{m=-\infty}^{\infty} x[m] \frac{1}{2\pi} \int_{2\pi} e^{j\omega(n-m)} d\omega.$$

- Orthogonality over one period: for integer $r = n - m$, complete cycles integrate to zero:

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j\omega r} d\omega = \begin{cases} 1 & r = 0 \\ 0 & r \neq 0 \end{cases} \implies \hat{x}[n] = x[n].$$

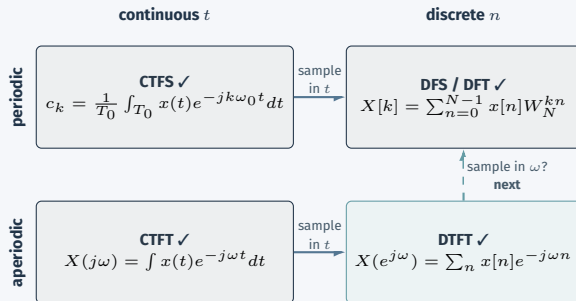
Inverse DTFT

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

DTFT Pair

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$
$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

- $x[n]$ is a continuum superposition of $e^{j\omega n}$ over one period – versus the DFS: a finite sum of N of them.



The DFT Through the DTFT Lens

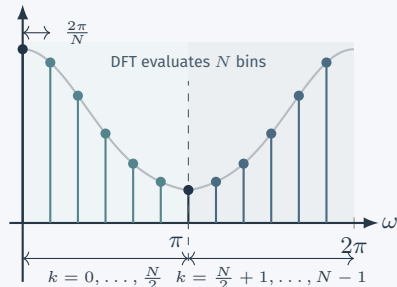
The DFT Samples the DTFT

- Just look at the DFT definition: it is the DTFT evaluated at **discrete frequencies** $\omega = \frac{2\pi}{N}k$ — nothing to derive:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn} = X(e^{j\omega}) \Big|_{\omega=\frac{2\pi}{N}k}.$$

- So the N -point DFT takes N **evenly spaced samples** over one 2π period of the DTFT.
- For even N : bins $0, \dots, \frac{N}{2}$ climb from dc to the highest positive frequency; the rest hold the folded negatives.

$$|X(e^{j\omega})|$$



One transform, two views

Periodic view: Fourier-series coefficients. Aperiodic view: N samples of the DTFT.
Same N numbers.

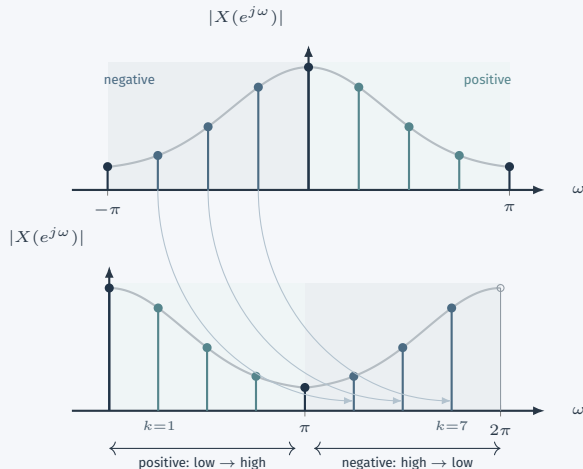
- Bin k sits at $\omega_k = \frac{2\pi}{N}k$; past $k = \frac{N}{2}$ these are the folded negative frequencies. For $N = 8$:

k	0	1	2	3	4	5	6	7
ω_k	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$-\frac{3\pi}{4}$	$-\frac{\pi}{2}$	$-\frac{\pi}{4}$

- dc $\rightarrow$ rising positive $\rightarrow$ highest (π) $\rightarrow$ negative, falling back toward dc.

How to read N bins

Bins $0 \rightarrow \frac{N}{2}$: **positive** frequencies, low $\rightarrow$ high. Bins $\frac{N}{2} \rightarrow N$: **negative** frequencies, high $\rightarrow$ low. A real lowpass signal peaks at $k = 0$ and near $k = N$, and dips at $k = \frac{N}{2}$.



Zero-Padding: Same Spectrum, Finer Samples

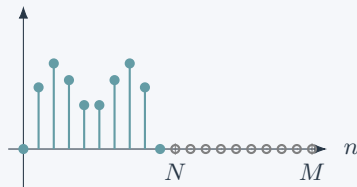
- **Zero-padding:** for a length- N sequence $x[n]$, artificially append zeros to reach a total length $M \geq N$.
- Appended zeros add nothing to the DTFT sum — the spectrum is unchanged. But the M -point DFT now samples it at spacing $\frac{2\pi}{M}$ instead of $\frac{2\pi}{N}$:

$$X_M[k] = \sum_{n=0}^{M-1} x[n] e^{-j\frac{2\pi}{M}kn} = X(e^{j\omega}) \Big|_{\omega=\frac{2\pi}{M}k}.$$

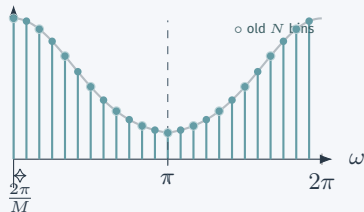
More pixels, not more information

Zero-padding interpolates the picture of the same DTFT. It cannot resolve what the N real samples never captured.

$x[n]$ padded



$|X(e^{j\omega})|$



A Complete Numeric Example ($N = 4$)

- Let $x[n] = \{1, 2, 3, 4\}$. With $N = 4$:
 $W_4 = e^{-j\frac{2\pi}{4}} = -j$, so $W_4^{kn} = (-j)^{kn}$ cycles through $1, -j, -1, j$.

- Apply $X[k] = \sum_{n=0}^3 x[n]W_4^{kn}$:

$$X[0] = 1 + 2 + 3 + 4 = 10$$

$$X[1] = 1 - 2j - 3 + 4j = -2 + 2j$$

$$X[2] = 1 - 2 + 3 - 4 = -2$$

$$X[3] = 1 + 2j - 3 - 4j = -2 - 2j$$

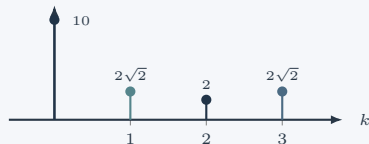
- Check the bins: $\omega_3 = \frac{2\pi \cdot 3}{4} = \frac{3\pi}{2} \equiv -\frac{\pi}{2}$, and indeed

$$X[3] = X^*[1].$$

👉 **Interactive:** this very $x[n]$, with every bin to hover

W_4^{kn}	$n=0$	$n=1$	$n=2$	$n=3$
$k=0$	1	1	1	1
$k=1$	1	$-j$	-1	j
$k=2$	1	-1	1	-1
$k=3$	1	j	-1	$-j$

$|X[k]|$



Negative-frequency partners

Bin 3 is bin 1's negative-frequency partner: for real signals $|X[k]|$ is symmetric about $\frac{N}{2}$. (Full symmetry properties: next lecture.)

Back to Physical Frequency

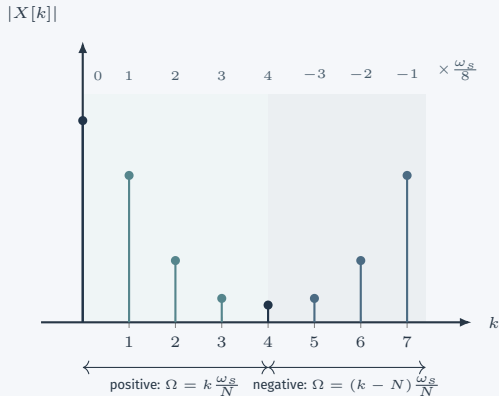
- The DTFT axis was normalized: $\omega = \Omega T_s$, where Ω is physical frequency (rad/s) and $\omega_s = \frac{2\pi}{T_s}$. Undo it for bin k :

$$\omega_k = \frac{2\pi}{N} k \mapsto \Omega_k = \frac{\omega_k}{T_s} = k \frac{\omega_s}{N}.$$

- The fold carries over in physical units: bins $0 \rightarrow \frac{N}{2}$ span $0 \rightarrow \frac{\omega_s}{2}$; bins $\frac{N}{2} \rightarrow N$ span $-\frac{\omega_s}{2} \rightarrow 0$.
- Relabeling the axis changes **units, not samples**: the same N numbers, the same fold.

π per sample = half the sampling rate

$\omega = \pi \leftrightarrow \Omega = \frac{\omega_s}{2}$: one 2π period of the DTFT is one ω_s of physical frequency, exactly the $[-\frac{\omega_s}{2}, \frac{\omega_s}{2}]$ window from the DTFT lecture.



Same stems as ever — only the ruler above them is new ($N = 8$).

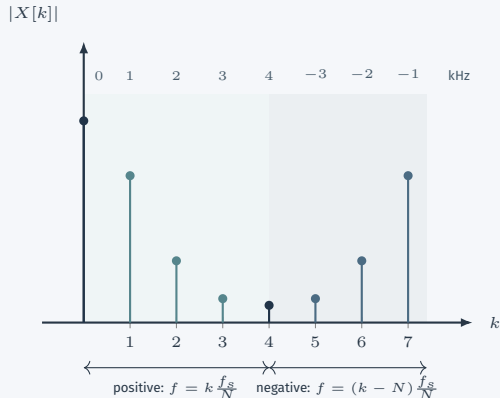
- The samples came from a clock: rate $f_s = \frac{1}{T_s}$. Relabel once more, to Hz:
 $f = \frac{\Omega}{2\pi}$, so $\frac{\omega_s}{2} \mapsto \frac{f_s}{2}$ and the bin spacing is

$$\Delta f = \frac{f_s}{N} = \frac{1}{NT_s} = \frac{1}{T_0}.$$

- Bin k corresponds to:

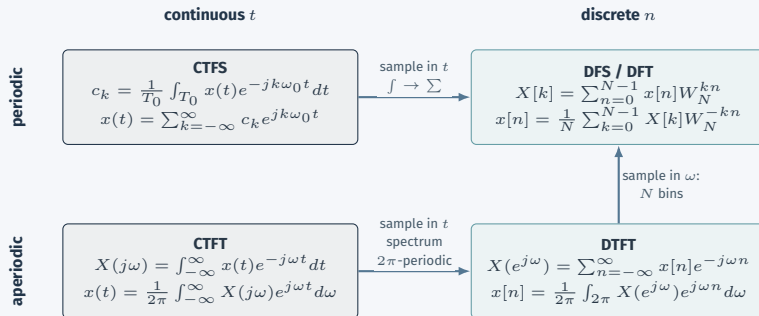
$$f_k = \begin{cases} k \frac{f_s}{N} & 0 \leq k \leq \frac{N}{2} \\ (k - N) \frac{f_s}{N} & \frac{N}{2} < k \leq N - 1 \end{cases}$$

- This is exactly the ordering `numpy.fft.fftfreq` reports, and why spectrum tools show 0 to $\frac{f_s}{2}$.



Example: $N = 8$ samples at $f_s = 8$ kHz $\implies \Delta f = 1$ kHz per bin.

The Big Picture



Discrete in one domain $\iff$ periodic in the other — proven, not promised

Sampling time made the spectrum 2π -periodic. Sampling frequency (keeping N bins) made time N -periodic.

DFS Pair ($\tilde{x}[n]$ is N -periodic)

$$a_k = \frac{1}{N} \sum_{n=0}^{N-1} \tilde{x}[n] e^{-j \frac{2\pi}{N} kn}$$
$$\tilde{x}[n] = \sum_{k=0}^{N-1} a_k e^{j \frac{2\pi}{N} kn}$$

DFT / IDFT Pair ($W_N = e^{-j2\pi/N}$)

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}$$
$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn}$$

DTFT Pair

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$
$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

The Three Bridges

- $c_k = \frac{1}{T_0} X[k]$ (sampled continuous periodic)
- $X[k] = N a_k$ (DFT = DFS, rescaled)
- $X[k] = X(e^{j\omega})|_{\omega=\frac{2\pi}{N}k}$ (DFT samples DTFT)

Reading the bins – one last time

Bins $0 \rightarrow \frac{N}{2}$: positive frequencies, low $\rightarrow$ high. Bins $\frac{N}{2} \rightarrow N$: negative frequencies, high $\rightarrow$ low.

Read

- Oppenheim, Willsky & Nawab, §5.1 — the DTFT, developed from the discrete Fourier series.
- Tan & Jiang, §4.1 — the DFT, developed from Fourier series coefficients of periodic digital signals.

Try

- Compute the 4-point DFT of $x[n] = \{1, 0, -1, 0\}$ by hand. Which bins light up, and why those two?
- In numpy: compare `fft(x)` of a signal with and without zero-padding.
- Play with the [interactive demos](#) linked at the bottom of the slides: one lab for periodic signals, one for aperiodic.

Next class

Properties of the DFT — linearity, time and frequency shifts, conjugate symmetry, Parseval's relation, and the convolution theorem. After that: why the DFT's N^2 cost is a scandal, and the fix that was found in 1805 — the FFT.